

## Chapter 103 — Physiology and Pharmacology of the Bladder and Urethra — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

- **M3 (muscarinic) contracts detrusor**; M2 more numerous (Gq odd M1/3/5; Gi even M2/4).
- **$\beta 3$  = dominant detrusor adrenoceptor** → relaxation via cAMP → mirabegron.
- **$\alpha 1A$  = urethra/prostate;  $\alpha 1D$  = storage symptoms** of bladder outlet obstruction (BOO).
- **Normal reflex = A $\delta$  (lightly myelinated) fibres, spinobulbospinal via pontine micturition centre (PMC); C fibres silent** until spinal cord injury (SCI)/inflammation.
- **Capsaicin/RTX (resiniferatoxin) block C-fibre reflex, spare A $\delta$ .**
- **Storage reflexes spinal; voiding needs the pons** (Barrington nucleus = PMC).
- **Pdet = Pves – Pabd** (detrusor = vesical – abdominal); **W = Pdet × Q** — low female voiding pressure ≠ weak detrusor.
- **Filling pressure <10 cmH<sub>2</sub>O**; compliance  $C = \Delta V / \Delta P$ .
- **Post-void residual (PVR) ≥300 mL** → evaluate underactive bladder.
- **OBTX (onabotulinumtoxinA) cleaves SNAP-25**; acts on efferents, afferents and urothelium.

### ANATOMY AND UROTHELIUM

- **Urothelium ~7 layers**; umbrella cells carry **uroplakins** (asymmetrical unit membrane, AUM, plaques).
- **E. coli type 1 fimbriae bind uroplakins.**
- **Tight junctions**: zona occludens-1 (ZO-1), occludin, claudin-4/-8/-12; **stretch → transepithelial resistance (TER) ↓ tenfold**, small-molecule permeability unchanged.
- **Glycosaminoglycan (GAG) layer**: antiadherence only; **not proven primary barrier**; main human GAG **chondroitin sulfate**.
- **Osmolality**: plasma ~300 vs urine 100–1500 mOsm/kg.
- **Water permeability** via aquaporins  $\approx 6.5 \times 10^{-5}$  cm/s.
- **Suburothelial myofibroblasts**: vimentin+,  $\alpha$ -SMA+, **desmin–**, Cx43-coupled.
- **Male bladder neck complete ring** → antegrade ejaculation; **female: no collar.**
- **Collagen I, III, IV**; poor compliance ↔ **↑ type III : I** and ↑ connective tissue : muscle.
- **Bladder wall ~50% collagen, 2% elastin.**

### URETHRA

- **Male urethra 4 segments**: preprostatic → prostatic → membranous → bulbous/penile.
- **Rhabdosphincter (external urethral sphincter, EUS) fibres**: **male 65% slow/35% fast; female 87% slow/13% fast.**
- **EUS cells 15–20  $\mu$ m**; distal slow-twitch sphincter  $\approx$  **50% static resistance.**
- **Nicotinic block ↓ tone ≤40%**;  **$\alpha$ -block ↓ urethral pressure ~½**; cholinergic negligible.
- **DeLancey hammock**: pressure compresses urethra onto rigid posterior support.
- **Female EUS horseshoe**; **male crescent → horseshoe (distal to verumontanum) → crescent.**

### SMOOTH MUSCLE

- **Filaments**: thick myosin 15 nm; thin actin 6–8 nm; intermediate desmin/vimentin 10 nm.
- **Bladder ~100% SM-B myosin** (faster ATPase); obstruction induces **SM-A.**
- **Sequence**:  $Ca^{2+}$  → calmodulin (CaM) → myosin light chain kinase (MLCK) → MLC20 (myosin light chain) phosphorylation → cross-bridge.
- **Calcium sensitisation**: RhoA-GTP → Rho kinase (ROK) → **myosin light chain phosphatase (MLCP) inactivated** ( $\pm$  CPI-17).
- **Resting potential –50 to –60 mV**;  $E_K \approx -90$ ,  $E_{Na} \approx +65$ ,  $E_{Cl} \approx -85$  mV.

- **AP**: phase 1 upstroke = **ICa (L-type)**; phases 2–3 = **IK (BK, large-conductance  $Ca^{2+}$ -activated  $K^+$ , most critical).**
- **L-type blockers**: nifedipine, diltiazem, verapamil; **T-type: no specific blocker.**
- **$Ca^{2+}$  sparks (ryanodine receptor) → BK → spontaneous transient outward currents → relaxation.** Store-operated entry = STIM + ORAI.
- **Connexin 43 (Cx43)**: 4 transmembrane domains; 6 = hemichannel; 2 hemichannels = channel.
- **Normal detrusor poorly coupled** — no fused tetanus.
- **Interstitial cells**: c-KIT (imatinib ↓ spontaneous activity); **PDGFR $\alpha$  (platelet-derived growth factor receptor- $\alpha$ ; P2Y1, inhibitory).**
- **Purinergic (atropine-resistant) contraction only in diseased human bladder** (obstruction, overactive bladder, ageing).

### BLADDER MECHANICS

- **Pdet at maximum flow: stress-incontinent (SUI) women ~19 vs healthy men 60–70 cmH<sub>2</sub>O.**
- **Low compliance causes**: fast fill, ↑ collagen/ ↓ elastin, muscle hyperactivity.
- **Pelvic nerve section ↓ compliance**; **acute cord transection does not.**
- **Male nomograms invalid in women.**

### NEURAL CONTROL

- **Parasympathetic S2–S4 → pelvic nerve**: contract detrusor, relax urethra (NO).
- **Sympathetic thoracolumbar → hypogastric**:  $\beta 3$  relax body,  $\alpha 1$  contract outlet.
- **Somatic Onuf's nucleus → pudendal**: EUS.
- **Postganglionic parasympathetic neurons in bladder wall → cauda equina** = decentralised, not denervated.
- **A $\delta$ /C cut-off**: 2 m/s cat, 1.3 m/s rat; **A $\delta$  ~30% of afferents.**
- **Mouse low-threshold muscular afferents 65–80%.**
- **TRPV1 >43 °C/protons**; **TRPV4 >27 °C/stretch**; **TRPM8 <23 °C/menthol**; **TRPA1 mustard oil, acrolein.**
- **RTX** from Euphorbia resinifera; isolated 1975; ultrapotent capsaicin analogue 1989.
- **Storage (Table 103.3)**: low afferent → EUS + internal sphincter contract, detrusor/ganglion inhibited — **spinal.**
- **Micturition**: high afferent → all reversed — **spinobulbospinal.**
- **Sphincter relaxes seconds before detrusor contracts.**
- **Barrington urethra-to-bladder**: pudendal → **PMC**; pelvic → **spinal.**
- **PMC does not project to Onuf's nucleus** — inhibits it via GABA/glycine interneurons.

- **L-region** = pontine continence/storage centre.
- **Pontine lesion** → retention; medial frontal/white matter → incontinence.
- **Dopamine inhibits PMC** — lost in Parkinson disease.
- **Rat spinal  $\alpha 1$ :  $\alpha 1A$  70%,  $\alpha 1B$  30%.**
- **Glutamate** = main excitatory transmitter (NMDA/AMPA); glycine/GABA inhibit.
- **7 P2X, 8 P2Y subtypes; P2X1 dominant in detrusor**,  $\uparrow$  in BOO.
- **PET = sustained states; fMRI = fast events; SPECT = poor resolution.**
- **Right prefrontal activation during storage** (PET positron emission tomography; fMRI functional MRI; SPECT single-photon emission CT).

## PHARMACOLOGY

- **Atropine abolishes normal human detrusor contraction.**
- **5 muscarinic subtypes; M3** → M2 shift in neurogenic/obstructed bladder.
- **Darifenacin  $\sim 30\times$  M3 > M2 (not tissue-selective); solifenacin/oxybutynin  $\sim 10\times$ .**
- **Tolterodine, solifenacin: bladder > salivary** — tissue selectivity matters more.
- **Desethyloxybutynin** → dry mouth → controlled-release oxybutynin better tolerated.
- **Mirabegron ( $\beta 3$ ) approved**; mirabegron + solifenacin effective.
- **Rat BOO  $\alpha 1D$  25% → 70%**; tamsulosin works,  $\alpha 1A$ -only antagonist does not.
- **Human urethra: no postjunctional  $\alpha 2$ .**
- **NK1** ← substance P (SP); **NK2** ← NKA; **NK3** ← NKB. Aprepitant (NK1)  $\downarrow$  OAB episodes.
- **Detrusor PG potency**: PGF2 $\alpha$  > PGE > PGE2; indomethacin  $\downarrow$  DO.
- **Endothelin 21 amino acids**; endothelin-A (ETA) antagonists  $\downarrow$  DO in BOO.
- **Oestrogen alone disappointing**; systemic may  $\uparrow$  UI.
- **Duloxetine (serotonin-noradrenaline reuptake inhibitor)** → Onuf's → EUS: approved Europe, not USA.

## CLINICAL RELEVANCE

- **Detrusor overactivity (DO) in only 11% of OAB.**
- **Mid-urethral sling (MUS) failure  $\sim 23\%$  at 1 year**;  $\uparrow$  urinary N-telopeptide (NTx) → more failure.
- **Spinal shock weeks** → neurogenic DO (NDO) via C-fibre spinal reflex.
- **After SCI**:  $\uparrow$  tetrodotoxin (TTX)-sensitive,  $\downarrow$  TTX-resistant Na<sup>+</sup> and A-type K<sup>+</sup> channels.
- **Anti-nerve growth factor (NGF) intrathecal  $\downarrow$  DO, detrusor-sphincter dyssynergia (DSD), autonomic dysreflexia** (rat).
- **Rat BOO**: bladder  $\times 15$  weight, surface  $\times 6$ , thickness  $\times 3$ , no mitoses.
- **Urinary NGF (pg/mL)**: control 0.5; BOO 1; BOO+OAB 41; BOO+DO 50.
- **Glomerulations**: Messing and Stamey 1978; specificity questioned.
- **Bladder pain syndrome/interstitial cystitis (BPS/IC)**: Europe = biopsy; USA = symptoms  $\pm$  hydrodistension.
- **Tanezumab failed in BPS/IC; RTX RCT negative; AF-219 (P2X3 antagonist) promising.**
- **Antiproliferative factor (APF) receptor CKAP4/p63**; frizzled 8 homology.
- **Nocturia**: Cx43 circadian —  $\downarrow$  during rest →  $\uparrow$  capacity; Cry knockout abolishes.

- **Ageing**:  $\downarrow$  contractility, sensation, urethral pressure; **capacity  $\downarrow$  only with DO.**
- **Aged bladder: noradrenaline (NA) contracts** ( $\uparrow \alpha$ ,  $\downarrow \beta$ ).
- **Pudendal vs sacral neuromodulation (SNM)**: 63% vs 46% improvement.
- **Pudendal 3 Hz inhibits; intermittent 20 Hz empties** (SCI cat).
- **OBTX 150 kDa**: heavy 100 + light 50 (zinc).
- **ABC (2012)**: OBTX = antimuscarinic for urgency incontinence (UUI); OBTX more complete dryness, more retention/UTI.
- **ROSETTA (2016)**: OBTX 200 U  $\approx$  SNM; OBTX more dry, more UTI.
- **FootStim 3 h daily**  $\downarrow$  UUI (refractory OAB).
- **Dates**: Barrington pons 1921/1925; Nobel GPCR 2012 (Kobilka, Lefkowitz); integral theory 1993; DHIC (detrusor hyperactivity with impaired contractility) 1987.

## CLASSIC TRAPS

- **Trap**: M2 mediates contraction (most numerous) — **Actually**: M3 contracts; M2 amplifies.
- **Trap**: M2 (Gi) raises cAMP — **Actually**: Gi inhibits adenylyl cyclase →  $\downarrow$  cAMP.
- **Trap**: low Pdet in a woman = weak detrusor — **Actually**: wide outlet;  $W = Pdet \times Q$ .
- **Trap**: GAG layer is the main barrier — **Actually**: uroplakins + tight junctions.
- **Trap**: ZO-1/claudins are gap-junction proteins — **Actually**: tight-junction; gap = connexins.
- **Trap**: PMC drives Onuf's nucleus directly — **Actually**: indirect GABA/glycine inhibition.
- **Trap**: ageing shrinks capacity — **Actually**: reduced capacity tracks DO, not age.
- **Trap**: selective  $\alpha 1A$  blocker improves LUTS — **Actually**:  $\uparrow$  flow only;  $\alpha 1D$  needed.
- **Trap**: sympathetic storage reflex vital in humans — **Actually**: retroperitoneal lymph node dissection leaves storage intact.
- **Trap**: ice-water test is clinical — **Actually**: research only; C-fibre reflex marker.